

- 3 (a) Scientists investigated the progress of reactions catalysed by two enzymes: dopa oxidase and neutrase. The reactions catalysed by these enzymes result in changes to the appearance of the reaction mixtures.

The reactions are shown in Fig. 3.1.

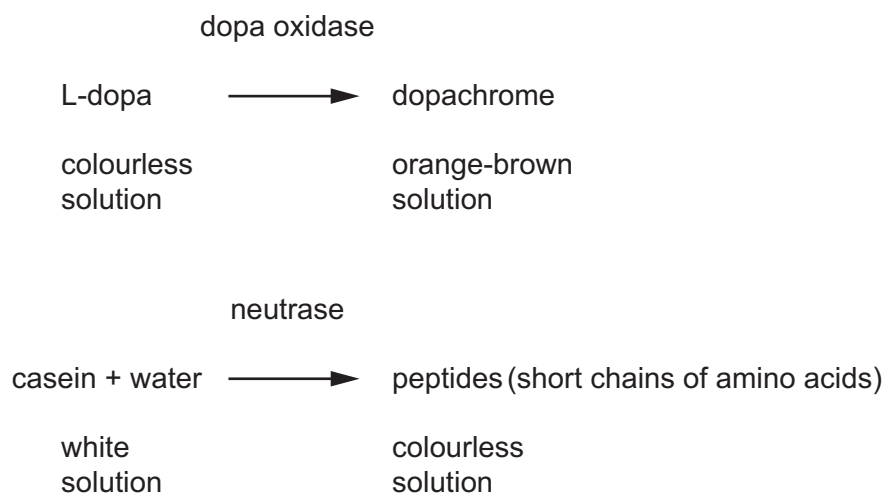


Fig. 3.1

The changes in appearance of the reaction mixtures make it possible to follow the reactions using a colorimeter.

Fig. 3.2 shows the progress of the reaction catalysed by dopa oxidase as recorded from a colorimeter.

Fig. 3.3 shows the progress of the reaction catalysed by neutrase as recorded from a colorimeter.

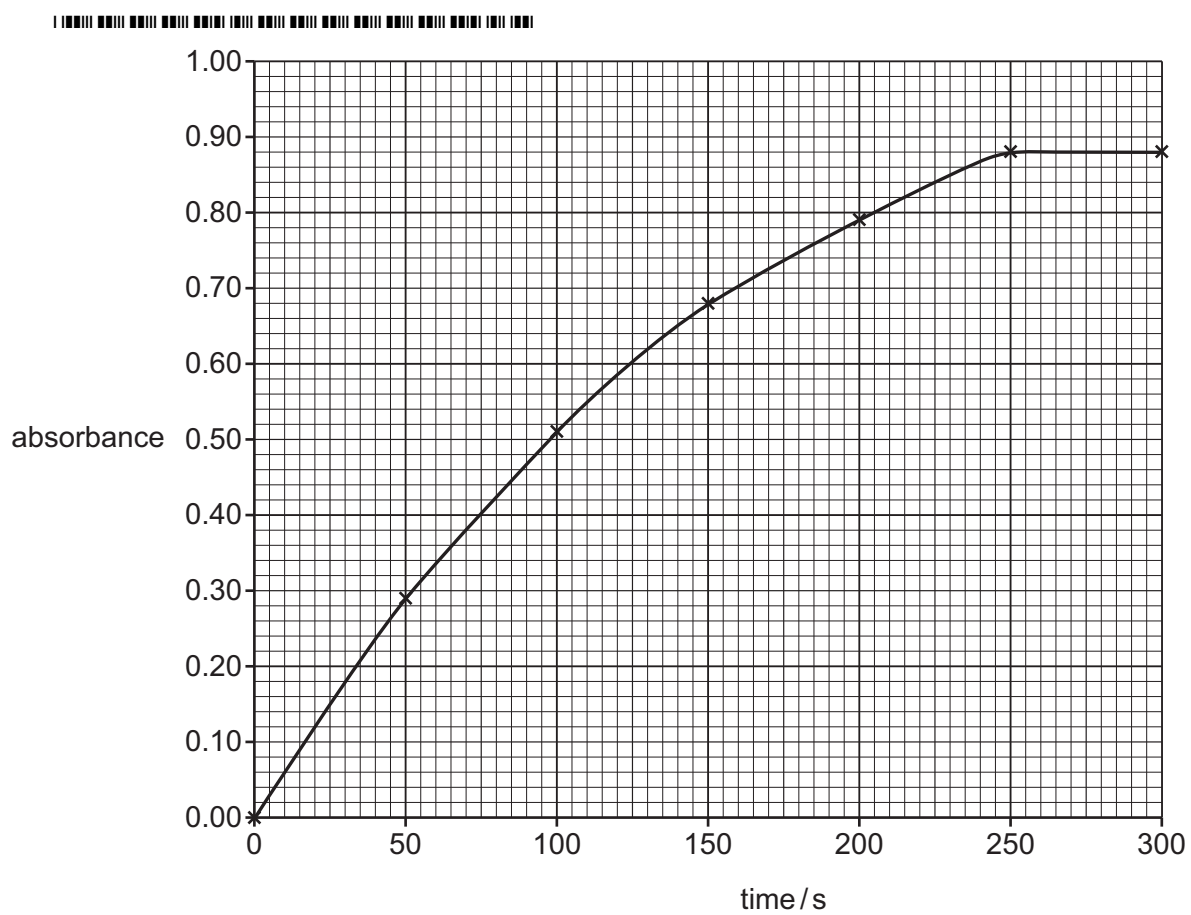


Fig. 3.2

- (ii) Suggest **two** advantages of using a colorimeter to investigate the progress of reactions such as those shown in Fig. 3.1.

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- (b) Scientists searching for a suitable enzyme to use in an industrial process isolated the bacterium *Vibrio parahaemolyticus* from the mouth of the Mediterranean eel, *Muraena helena*.

The scientists discovered an enzyme in the bacterium that was suitable for the industrial process. The scientists named the enzyme VpSP37.

The scientists investigated how the rate of reaction catalysed by VpSP37 is affected by the concentration of its substrate. The results of the investigation are shown in Fig. 3.4.

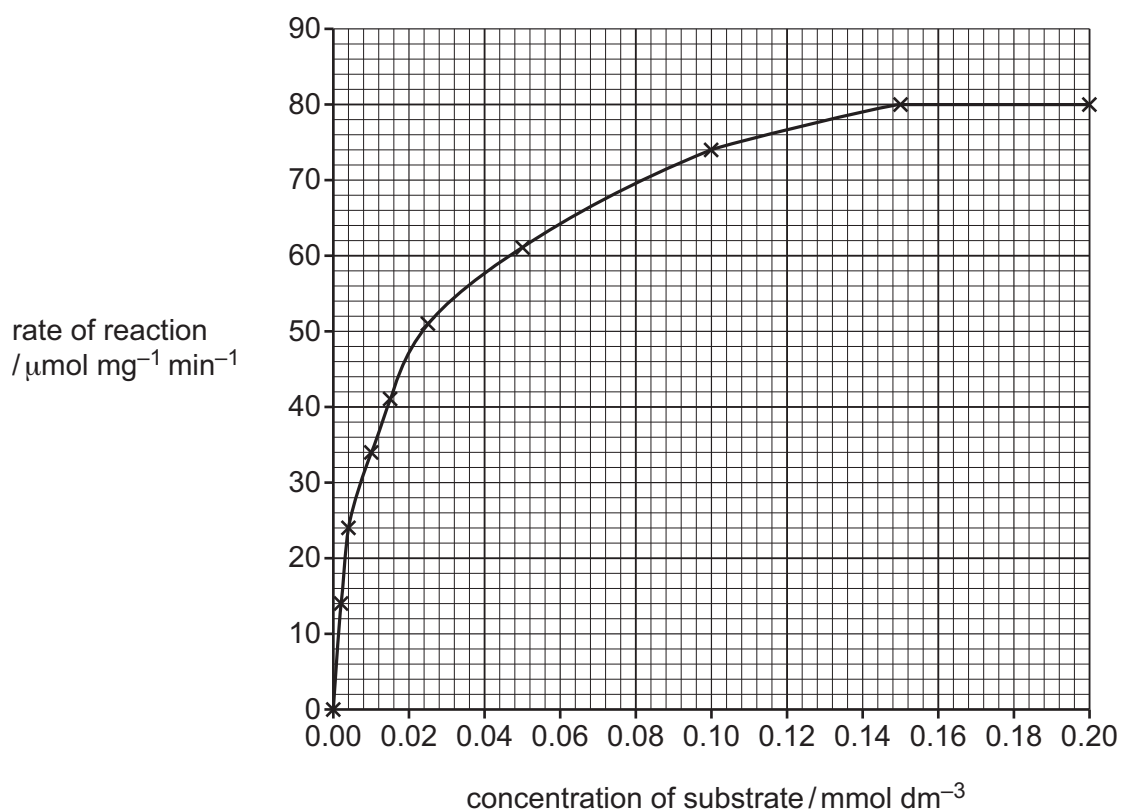


Fig. 3.4



- (i) Calculate the Michaelis–Menten constant, K_m , for the enzyme VpSP37 using the information in Fig. 3.4.

Show your working.

$K_m = \dots\dots\dots$ [2]

- (ii) The scientists discovered other enzymes that were suitable for the industrial process. These enzymes had higher K_m values than VpSP37.

Explain the advantage of using the enzyme VpSP37 in the industrial process rather than one of these other enzymes with higher K_m values.

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[Total: 9]